BRAMHALL COLLIERY - 400 m LEVEL VENTILATION - DATA SHEET (sheet 1 of 3)

Arrangement. Fresh air enters through the downcast shaft DS to the pit bottom PB and returns through the upcast shaft US from the upcast bottom UB. The surface exhaust fan (main fan, with a standby fan on the same drift) draws air up US. The airways of the 400 m level are shown on sheet 3 (ventilation plan): intake airways I1-I4, W1, C1, E1; return airways W2, C2, E2, T1, R1-R3; crosscuts X1 (J7 to J8) and X2 (J8 to J9). Each airway consists of the sections shown on the plan, in the order shown, separated at survey stations (SS) where the roadway profile changes. The working faces of the W, C and E panels are in airways W2, C2 and E2: face airflow = airflow in that airway.

Normal arrangement. Crosscut X1 is closed by door pair D1 (shut: no airflow; open: no added resistance). Crosscut X2 is sealed by stopping ST-7 (intact: no airflow). Regulators REG-W, REG-C, REG-E and REG-T are in airways W2, C2, E2 and T1. Normal regulator openings: REG-W 2.4 m2, REG-C 1.8 m2, REG-E 2.8 m2, REG-T 1.5 m2. Main fan blade setting 0 deg.

Air. Density rho = 1.20 kg/m3 throughout, incompressible flow; natural ventilation pressure is neglected (both shaft tops are at the same atmospheric pressure).

Roadway profiles (Atkinson friction factor k at 1.20 kg/m3):

P1 arched main, steel arches, 5.2 m span: area 16.0 m2, perimeter 15.2 m, k = 0.0090 kg/m3
P2 rectangular, roof bolted, 4.8 x 3.0 m: area 14.4 m2, perimeter 15.6 m, k = 0.0110 kg/m3
P3 rectangular, roof bolted, 4.4 x 2.6 m: area 11.4 m2, perimeter 14.0 m, k = 0.0130 kg/m3
P4 panel gate, yielding arches, 3.8 x 2.5 m: area 9.5 m2, perimeter 12.6 m, k = 0.0170 kg/m3
P5 return, rectangular, 4.6 x 2.6 m: area 12.0 m2, perimeter 14.4 m, k = 0.0150 kg/m3

Formulas.

Airway section resistance R = k L per / A^3 (N s2/m8), L section length (m), per perimeter (m), A area (m2).
Airway pressure drop p = R Q |Q| (Pa), Q airflow (m3/s); R of an airway = sum of its section resistances plus any regulator or brattice in it.
Regulator resistance R_reg = rho / (2 (Cc A_o)^2), Cc = 0.65, A_o = regulator opening (m2).
A damaged stopping replaced by a temporary brattice: resistance 3.5 N s2/m8 in series with the crosscut's sections.
Shafts: downcast DS R = 0.0042 N s2/m8 (surface to PB); upcast US including fan drift R = 0.0048 N s2/m8 (UB to fan).
Fan static pressure p_fan = a - b1 Q - b2 Q^2 (Pa, Q in m3/s) for the fan in service:
main -5 a = 3050 Pa, b1 = 2.0 Pa s/m3, b2 = 0.047 Pa s2/m6
main 0 a = 3300 Pa, b1 = 2.0 Pa s/m3, b2 = 0.044 Pa s2/m6
main +5 a = 3550 Pa, b1 = 2.0 Pa s/m3, b2 = 0.042 Pa s2/m6
standby a = 3050 Pa, b1 = 1.2 Pa s/m3, b2 = 0.049 Pa s2/m6
The fan pressure balances the total pressure drop surface - DS - level - UB - US - fan. Static pressure at a node is reported relative to surface atmospheric pressure (Pa, negative below atmospheric); so p(PB) = -R_DS Q_fan^2 and p(UB) = -(p_fan - R_US Q_fan^2).

Statutory minimum airflow at every working face: 30.0 m3/s.

BRAMHALL COLLIERY - VENTILATION CONTROL ROOM LOG, NIGHT SHIFT (sheet 2 of 3)

22:00 Handover from back shift: main fan running, blade setting 0 deg. Regulators at normal openings. Door D1 shut, stopping ST-7 intact.
00:30 T1 heading stopped for the night. REG-T closed down from 1.5 to 1.0 m2.
01:00 Manager requests main fan blade setting +5 deg for the morning; change to be made by the day shift fitters (not done on nights).
02:00 Methane 1.1 % in E panel return. REG-E opened from 2.8 to 3.6 m2.
03:20 Door D1 pair propped open for the shuttle car transfer from W panel to C panel.
04:10 Main fan tripped on bearing temperature. Standby fan started automatically 04:11 and is in service.
05:10 Roof fall in crosscut X2 damaged stopping ST-7; air passing through X2.
05:45 REG-C opening checked: 1.8 m2 as set.
06:20 Temporary brattice erected at ST-7 (stopping not rebuilt).
06:30 Main fan bearing replaced. Main fan held on standby pending vibration test by day shift; standby fan remains in service.
07:15 Door D1 still propped open, shuttle car transfer continuing; leave open until day shift.
07:40 Methane E return 0.8 %. REG-E to be reset to 2.8 m2 at end of shift.
07:55 REG-E reset deferred to day shift (methane still above 0.75 %); REG-E remains at 3.6 m2.
08:00 Handover to day shift.

BRAMHALL COLLIERY - sheet 3: ventilation plan, 400 m level (schematic, NOT to scale; lengths as labelled). Each airway is shown section by section with roadway profile and section length; profile changes are bars with survey station numbers SSn. Green: intake airways; brown: return airways; violet: crosscuts. Yellow squares: regulators REG, door D1, stopping ST-7.

